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E-commerce integration, financial services enhancement, and rural revitalization: A quasi-natural experiment based on the comprehensive demonstration county pilot of e-commerce in rural areas

Yifan Qi [a], Yasi Gao [b], Yanghui Wu [c,*]

[a] China School of Sociology, Beijing University of Technology, Beijing 100124, China
[b] Business Intelligence College, Zhejiang Institute of Economics and Trade, Hangzhou 310018, China
[c] Weifang Vocational College, Weifang, Shandong 261041, China

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ABSTRACT

This paper, based on provincial panel data from China from 2007 to 2023, employs a combination of benchmark regression, Difference-in-Differences (DID), and mediation effect models to systematically analyze the impact mechanisms and pathways of the integration of e-commerce into rural areas and the enhancement of financial services on rural revitalization(RUR). Empirical results indicate that: (1) the integration of e-commerce into rural areas significantly promotes RUR, with its positive effects being more pronounced in regions with lower levels of fiscal support, highlighting significant regional heterogeneity; (2) the enhancement of financial services plays a robust and significant positive role in RUR; (3) the enhancement of financial services serves a crucial mediating role in the process by which e-commerce integration into rural areas affects RUR.

1. Introduction

1.1. Research background

Against the backdrop of an accelerating reshaping of the global economic landscape and deep adjustments in the domestic economic structure, rural economic development has become an important lever for promoting high-quality growth in China's economy and achieving coordinated urban-rural balance. Data from the National Bureau of Statistics indicates that by 2024, the actual growth rate of per capita disposable income for rural residents continues to outpace that of urban residents, while the level of rural infrastructure and public services steadily improves, and the industrial structure in rural areas gradually optimizes. However, rural development still faces prominent issues such as low industrial added value, insufficient resource allocation efficiency, and obstructed market access channels, necessitating the infusion of new development momentum and innovative models.

In this context, the advancement of e-commerce in rural areas (ECIR) and improvements in financial services have emerged as two key driving forces for current rural revitalization (RUR). With its distinctive advantages of overcoming geographical limitations,

* Corresponding author.
E-mail address: wyh309805228@126.com (Y. Wu).

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reducing transaction costs, and expanding market space, e-commerce has facilitated the upward movement of agricultural products, the upgrading of rural consumption, and the deep integration of rural industries. By the end of 2024, national rural online retail sales have exceeded 3 trillion yuan, marking a year-on-year growth of 15 %, and rural e-commerce is becoming an important engine for rural economic growth. Meanwhile, finance, as the core of the modern economy, provides solid support for rural revitalization through multiple functions, such as optimizing resource allocation, offering funding support, and diversifying operational risks.

1.2. Literature review

Research on the relationship between ECIR and RUR generally agrees that e-commerce can broaden the sales channels for agricultural products, enhance their added value, and promote rural employment and entrepreneurship, thereby driving the revitalization of rural industries and increasing farmers' incomes (Xu and Yang, 2025). For instance, relevant case analyses indicate that the construction of rural e-commerce platforms effectively alleviates the "difficulties in selling" agricultural products and improves farmers' economic returns (Chen and Long, 2024). However, most existing studies focus on specific regions or case analyses, lacking empirical tests based on large national samples, and rarely explore the heterogeneity of the impact of e-commerce on RUR.

In terms of enhancing financial services and rural revitalization, scholars widely recognize the positive role of financial services in the development of rural industries, improvement of infrastructure, and elevation of farmers' living standards (Liu and Wang, 2023). Some studies have verified the positive effects of financial development on rural economic growth through econometric models (Li and Zhang, 2024). Nevertheless, the existing literature has paid little attention to the mediating effects of financial service enhancement in the relationship between ECIR and RUR, and empirical research on the structural reform of the supply side of rural financial services is also insufficient.

In summary, current research on the relationships among ECIR, enhancement of financial services, and rural revitalization primarily focuses on pairwise relationships, and systematic exploration of their interlinked mechanisms is scarce. Additionally, in terms of research methods, there is a lack of empirical tests based on national provincial panel data and multiple models, resulting in the extrapolation and accuracy of existing conclusions needing further improvement.

1.3. Contribution

The main objective of this study is to systematically analyze the inherent connections and mechanisms among the incorporation of e-commerce into rural areas, the enhancement of financial services, and RUR. Through rigorous empirical analysis, the study aims to uncover the pathways and effects of e-commerce and financial services in the process of rural revitalization. The significance of this research lies in two aspects: first, it supplements and enriches the academic achievements in the field of the collaborative development of green finance, financial services, and rural revitalization, expanding the analytical perspective on the interaction mechanisms among the three. Second, the research findings provide solid quantitative evidence for policymakers to develop precise measures, which aids in optimizing the e-commerce initiatives in rural areas, strengthening the supply of financial services, promoting the synergy between e-commerce and finance, enhancing the effectiveness of RUR strategies, and continuously driving the high-quality development of the rural economy and the improvement of farmers' living standards.

2. Research hypotheses

With the rapid development of information technology, e-commerce has become an important driving force for the transformation and upgrading of the rural economy (Yu et al., 2024). Theoretically, the efficient allocation of information flow, capital flow, and commodity flow can promote the optimal movement of resource factors between urban and rural areas, thus fostering coordinated regional economic development (Gao, 2022). By accessing e-commerce platforms, rural areas can expand their market space, effectively broaden sales channels for agricultural products, enhance product added value, and significantly shorten the production and sales chain, thereby reducing transaction costs (Zhang et al., 2025). The entry of e-commerce into rural areas has facilitated the improvement of the information literacy level in rural regions, strengthening the capability of rural market entities to access information, which optimizes agricultural production decisions and enhances farmers' bargaining power (Zhang et al., 2022). The advancement of ECIR also drives the transformation of traditional agriculture towards the integration of multiple industries, stimulating the innovative vitality and developmental potential of rural regions, and providing strong support for the realization of the rural revitalization strategy.

H1. There is a significant positive correlation between the incorporation of ECIR and RUR.

Government financial support, as an important external institutional supply, plays a crucial role in rural infrastructure construction, industrial environment optimization, and risk mitigation (Zhan et al., 2025). In regions with high government financial support, substantial financial investment can significantly improve infrastructure such as transportation, logistics, and telecommunications, thereby laying a solid hardware foundation for the development of e-commerce (Qiu et al., 2024). At the same time, policies such as tax incentives and special subsidies can attract more e-commerce enterprises and talent, fostering industrial agglomeration. However, according to the theory of "crowding out effect," excessive fiscal intervention may suppress private investment willingness and weaken the innovation drive of market entities (Karine, 2021). In contrast, in regions with relatively low fiscal support, limited public resources compel local entities to stimulate market vitality and actively explore e-commerce development models with local characteristics. Farmers and enterprises, through combining local characteristic resources with e-commerce platforms, form differentiated

competitive advantages, effectively promoting rural revitalization (Wei et al., 2024).

H2. The positive impact of e-commerce on rural revitalization exhibits heterogeneity between regions of high government financial support and those of low government financial support.

From the perspective of capital allocation, efficient financial services can guide capital to flow toward high-return industries and projects in rural areas (Xu, 2022). In the process of rural economic development, traditional industry upgrades, the incubation of emerging rural industries, and infrastructure construction all urgently require substantial financial support (Su et al., 2024). The enhancement of financial services is typically manifested as an increase in the value-added ratio of the financial industry to GDP, indicating that financial institutions can more accurately identify and select quality projects, achieving effective capital transfer from inefficient sectors to efficient ones (Li and Gan, 2025). For instance, through innovative credit products, financial institutions can offer medium to long-term, low-interest loan support to rural e-commerce enterprises, assisting them in expanding their operational scale and improving supply chain systems, ultimately enhancing agricultural product sales and farmers' income, thereby promoting the prosperity of rural industries (Basnayake et al., 2024).

H3. The enhancement of financial services significantly promotes the level of rural revitalization.

Based on transaction cost theory, the initial stage of e-commerce entering rural areas often faces significant funding demands for infrastructure construction, supply chain integration, and market development (Hao et al., 2023). Due to widespread issues in rural areas such as the lack of collateral and an inadequate credit system, relying solely on the self-owned funds of market entities makes it difficult to achieve scaled development (Tian et al., 2023). At this point, the enhancement of financial services can effectively reduce transaction costs: on one hand, financial institutions widen financing channels for e-commerce enterprises through innovative products such as supply chain finance and intellectual property pledges, alleviating funding constraints (Luo et al., 2023); on the other hand, leveraging financial technology to optimize processes such as payment settlements and credit assessments reduces information asymmetry and transactional friction (Wei et al., 2025). Ultimately, this mechanism promotes rural industrial revitalization and increases farmers' incomes, forming a positive transmission chain of "e-commerce development – financial service support – rural revitalization advancement."

H4. The enhancement of financial services plays a mediating role in the relationship between e-commerce entering rural areas and rural revitalization.

3. Model and data sources

3.1. Model setting

3.1.1. Benchmark regression analysis model setting

This study establishes model (1) and model (2) to analyze the impacts of ECIR and the enhancement of financial services on rural revitalization, respectively. Both models introduce a series of control variables X, denoted as follows, mainly including financial agglomeration (FIA), government fiscal support intensity (Gup), education investment level (EduInv), industrial structure upgrading (Upgrade), urbanization level (UR), infrastructure construction (Infrastructure), economic development level (ECO), and urban residents' income level (cdev) as control variables. Regional effects (Regional) and time effects (Year) are fixed, and a random error term (ω_{it})is added to ensure a more comprehensive analysis of the relationships among the three.

$$RUR_{it} = \alpha + \alpha_1 ECIR_{it} + \alpha_2 X_{it} + Regional + Year + \omega_{it} \tag{1}$$

$$RUR_{it} = \beta + \beta_1 FS_{it} + \beta_2 X_{it} + Regional + Year + \omega_{it} \tag{2}$$

The dependent variable, RUR, denotes the level of rural revitalization and is constructed using a comprehensive evaluation index system encompassing five key dimensions: industrial prosperity, ecological livability, rural culture, effective governance, and prosperous living standards, following the framework proposed by Yang et al. (2021). Specifically, industrial prosperity is evaluated through indicators such as the growth rate of the total output value from agriculture, forestry, animal husbandry, and fishery, as well as the share of business income in the per capita disposable income of rural residents. Ecological livability is assessed by metrics including the treatment rate of rural domestic waste, the prevalence of sanitary toilets, and forest coverage in rural areas. Rural culture is measured by the number of cultural stations per 10,000 inhabitants and rural education investment. Effective governance encompasses the participation rate in grassroots democratic elections and the rate of public security incidents. Lastly, affluent living is reflected by both the per capita disposable income and the Engel coefficient of rural households. All indicators are assigned weights using the entropy method to derive a comprehensive score, where a higher value signifies greater achievements in rural revitalization.

The independent variable ECIR refers to the promotion of e-commerce in rural areas, based on relevant policies for this initiative. The comprehensive demonstration counties for ECIR are key national initiatives aimed at fostering rural e-commerce development and boosting local economic growth. Under this project, the government selects and provides financial support to representative regions across the country in order to expedite the advancement of local e-commerce industries. For empirical analysis, if a county is identified as a comprehensive demonstration county for ECIR during the policy implementation period, it is assigned a value of 1 in the treatment group; otherwise, counties are categorized in the control group and assigned a value of 0.

The independent variable FS denotes the enhancement of financial services, represented by the proportion of the added value of the

financial industry to GDP (Jiakui et al., 2023). Data on the added value of the financial industry is sourced from the "China Financial Statistical Yearbook." This indicator reflects the significance and development vitality of the financial sector in the regional economy; a higher proportion signifies a more significant enhancement of financial services. Additionally, in subsequent models, it serves as a mediating variable.

In addition, this study introduces the regional financial development level (FDL) as an instrumental variable, measured by the ratio of total deposits and loans of regional financial institutions to GDP. This indicator comprehensively reflects the scale and activity level of the regional financial system and is an important metric for assessing the maturity and development status of the regional financial system. The choice of this as an instrumental variable is justified for two reasons. First, the level of regional financial development is closely correlated with the enhancement of local financial services; the larger the deposits and loans of financial institutions, the more conducive it is to the optimization and upgrading of financial services. Second, based on existing research and data characteristics, this indicator does not have a direct correlation with other confounding factors in the rural revitalization process, aside from the enhancement channels of financial services, thus satisfying the requirements for the relevance and exogeneity of the instrumental variable.

3.1.2. Mediation effect model specification

To examine whether the enhancement of financial services acts as a mediating role between ECIR and RUR, this study constructs models (3) and (4):

$$FS_{it} = \gamma + \gamma_1 ECIR_{it} + \gamma_2 X_{it} + Regional + Year + \omega_{it} \tag{3}$$

$$RUR_{it} = \mu + \mu_1 ECIR_{it} + \mu_2 FS_{it} + \mu_k X_{it} + Regional + Year + \omega_{it} \tag{4}$$

3.2. Summary statistics of variables

This study utilizes authoritative databases such as the China Statistical Yearbook from 2007 to 2023 as data sources, selecting relevant panel data from 31 provinces. During the data preparation process, this paper removed outliers from the samples and supplemented missing key values with reference to regional statistical yearbooks for correction. Outliers and missing values could introduce bias in the model's estimation results; therefore, they were eliminated during data processing to ensure the accuracy of the data and the validity of the analysis. In Table 1, the mean of RUR is 0.3831, indicating significant disparities in the level of rural revitalization among the sample regions, as well as a pronounced imbalance in rural revitalization development. The mean for ECIR is 0.5028, suggesting that approximately half of the sample regions are rated as comprehensive demonstration counties for ECIR, with a noticeable binary distribution characteristic in policy coverage. The mean for financial service enhancement (FS) is 0.0669, indicating that the importance of financial services in the economy varies only slightly across regions, yet the overall level of financial service enhancement still requires improvement. Control variables encompass various aspects such as financial clustering and government fiscal support, with diverse ranges and degrees of dispersion that reflect rich differences among the sample regions in terms of economic, social, and policy environments, effectively controlling the relationship between core variables.

4. Empirical results and analysis

4.1. ECIR and rural revitalization

4.1.1. Benchmark regression analysis

Table 2 reports the impact of ECIR on rural revitalization under four scenarios: without control variables, with control variables, regional fixed effects, and double fixed effects. The results indicate that when control variables are not included, the regression coefficient for ECIR is significantly negative; however, the model's goodness of fit is only 0.0019, suggesting limited explanatory power. After introducing control variables, the coefficients for ECIR in columns (2) to (4) are all positive, indicating that after controlling for

Table 1
Descriptive statistical analysis.

VarName	Obs	Mean	SD	Min	Median	Max
RUR	527	0.3831	0.3242	0.0000	0.3255	1.7300
ECIR	527	0.5028	0.5005	0.0000	1.0000	1.0000
FS	527	0.0669	0.0319	0.0175	0.0632	0.1971
FDL	527	3.3389	1.1521	1.4535	3.0838	8.1641
FIA	527	0.9977	0.4331	0.3901	0.9008	3.0097
GUP	527	0.2758	0.1965	0.0969	0.2270	1.3538
EduInv	527	0.1627	0.0272	0.0989	0.1649	0.2244
Upgrade	527	0.8982	0.0542	0.7128	0.8994	0.9980
UR	527	0.5715	0.1411	0.2145	0.5673	0.8958
Infrastructure	527	24.2616	0.7235	22.0685	24.3172	25.9648
ECO	527	10.6897	0.5992	8.9592	10.7250	12.2075
cdev	527	10.2178	0.4719	9.2477	10.2695	11.4017

Table 2
Analysis results of the impact of ECIR on rural revitalization.

VARIABLES	(1) RUR	(2) RUR	(3) RUR	(4) RUR
ECIR	−0.1924***	0.1071***	0.1412***	0.1509***
	(−3.3450)	(2.8809)	(3.2015)	(3.4934)
FIA		0.0156	0.0017	0.0089
		(0.7649)	(0.0722)	(0.2855)
GUP		−0.4529***	−1.0225***	−0.3086*
		(−4.8403)	(−7.0330)	(−1.8732)
EduInv		0.7983*	−0.0863	1.3319***
		(1.8438)	(−0.1853)	(2.6593)
Upgrade		1.1665***	1.7374***	2.0728***
		(3.7846)	(3.8605)	(5.0053)
UR		0.0364	0.9171***	0.1882
		(0.1621)	(2.8537)	(0.5999)
Infrastructure		−0.0724***	0.1095**	0.1238**
		(−2.7972)	(2.1195)	(2.4725)
ECO		−0.0823	−0.3426***	−0.1700**
		(−1.0962)	(−4.0475)	(−1.9955)
Cdev		0.6678***	0.6819***	−0.0100
		(8.7413)	(8.0521)	(−0.0447)
Constant	0.1911***	−4.7794***	−7.2278***	−3.0538*
	(2.9418)	(−10.7176)	(−9.0160)	(−1.7133)
Regional FE	NO	NO	YES	YES
Year FE	NO	NO	NO	YES
N	527	527	527	527
R-squared	0.0019	0.7887	0.8042	0.8487

other influencing factors, ECIR effectively promotes the enhancement of rural revitalization levels. Moreover, the goodness of fit for these three sets of models exceeds 0.7887, demonstrating a good model fit, which strongly supports Hypothesis 1. Further comparison of different models shows that the double fixed effects regression not only achieves optimal significance but also has the best goodness of fit, with the significant coefficient for ECIR on RUR at 0.1509, and the highest goodness of fit among the four models. In conclusion, the selection of a model that includes all control variables and double fixed effects for regions and time is more suitable for subsequent analysis.

4.1.2. Propensity score matching and difference-in-differences estimations

In Table 3, the results of the parallel trend test indicated that, in the first five years following the implementation of ECIR policy, there were no significant findings across various regions. However, after the policy was implemented, the regression coefficients became significant, suggesting that the pilot policy for ECIR significantly promoted rural revitalization and confirmed the robustness of the regression analysis. At the same time, the results of the PSM-DID test indicated that the regression coefficient for ECIR was 0.1120, demonstrating that this policy can significantly enhance the level of rural revitalization. In this case, both the sign of the regression coefficient and its significance are consistent with the baseline regression results, further validating the robustness of the conclusions drawn from Model (1).

4.1.3. Heterogeneity analysis

To further reveal the heterogeneous impact of the e-commerce policy for rural areas on rural revitalization, this study divides the samples into two groups based on the intensity of regional fiscal support: high fiscal support and low fiscal support, and conducts empirical analysis on each group. The results are shown in Table 4. In regions with high fiscal support, the impact of the e-commerce policy on rural revitalization is not significant. However, in regions with low fiscal support, the regression coefficient of the policy is 0.1335, which is significant at the 1 % level, indicating that the e-commerce policy has a significant promoting effect on rural revitalization in areas with relatively limited fiscal support, thereby validating Hypothesis 2.

4.2. Financial services enhancement and rural revitalization

4.2.1. Benchmark regression analysis

Table 5 reports the results of the impact of financial services enhancement on rural revitalization under five model specifications, which sequentially incorporate control variables and dual fixed effects. The results indicate that the coefficient for financial services enhancement remains positive and statistically significant, suggesting that enhancing financial services can significantly promote rural revitalization, thus supporting Hypothesis 3. Particularly under the dual fixed effects model, the coefficient for financial services enhancement is highly significant and positive, with the best model fit, which validates the reasonableness of Model (2) specification.

4.2.2. Endogeneity test

This study selects regional financial development level (FDL) as an instrumental variable for analysis. In Table 6, FDL has a

Table 3
Parallel trend tests and PSM-DID.

Variables	Parallel trend test RUR	PSM-DID RUR
pre_5	0.0175	
	(0.4073)	
pre_4	0.0078	
	(0.2023)	
pre_3	−0.0065	
	(−0.1864)	
pre_2	−0.0107	
	(−0.3830)	
current	0.0090	
	(0.2480)	
post_1	0.0784**	
	(2.2098)	
post_2	0.0810**	
	(2.4243)	
post_3	0.1027***	
	(3.4123)	
post_4	0.1427***	
	(6.5239)	
ECIR		0.1120***
		(2.8039)
FIA	0.0064	0.0056
	(0.2416)	(0.1071)
GUP	−0.4152**	−0.0230
	(−2.2207)	(−0.1159)
EduInv	1.2756**	3.4200***
	(2.1346)	(4.8202)
Upgrade	2.0170***	1.7985***
	(5.7611)	(2.9029)
UR	0.2255	0.0386
	(0.4722)	(0.0653)
Infrastructure	0.1390**	−0.0110
	(2.5400)	(−0.1574)
ECO	−0.1415	−0.0915
	(−1.4024)	(−0.6367)
cdev	−0.1445	−0.1031
	(−0.5308)	(−0.3255)
Constant	−2.0383	0.0788
	(−0.7958)	(0.0338)
Regional FE	YES	YES
Year FE	YES	YES
N	527	221
R-squared	0.9028	0.8653

significant positive effect on financial service enhancement, and the fitted financial service enhancement exhibits a significant positive influence on RUR, consistent with the results of the baseline regression. The results suggest that the endogeneity problem has been successfully mitigated using the instrumental variable approach. Additionally, the LM statistic and the F statistic values confirm the validity of the selected instrument, and the regression model examining the influence of financial service improvements on rural revitalization demonstrates strong statistical reliability, supporting the validity of the conclusions.

4.2.3. Analysis of the mediating role of financial service enhancement

In Table 7, the findings indicate that ECIR has a significant positive effect on the enhancement of financial services. Additionally, both ECIR and the enhancement of financial services have a significant positive impact on RUR. This result suggests that financial service enhancement plays an important mediating role in the relationship between the integration of e-commerce into rural areas and rural revitalization, thereby confirming Hypothesis 4.

The Bootstrap test results presented in Table 8 indicate that the enhancement of financial services plays a significant mediating role between ECIR and rural revitalization. The coefficient for the mediating effect is 0.0695, and the 95 % confidence interval does not include zero, suggesting that this mediating effect is statistically significant. This confirms that the mediating effect of promoting RUR through the enhancement of financial services in the e-commerce integration policy is indeed present and robust.

5. Conclusions

Based on provincial panel data from China spanning 2007 to 2023, this study draws the following main conclusions: First, the integration of ECIR significantly promotes rural revitalization, particularly in regions with relatively low fiscal support, where its

Table 4
Heterogeneity analysis results table.

VARIABLES	(1) High financial support intensity	(2) Low financial support intensity
ECIR	0.0535	0.1335***
	(1.0828)	(3.7523)
FIA	0.0167	0.0516
	(0.4747)	(0.9272)
GUP	−0.1171	−2.8143**
	(−0.7675)	(−2.5449)
EduInv	0.6764	−0.4089
	(1.2552)	(−0.4122)
Upgrade	2.7257***	1.4770
	(6.4353)	(1.1287)
UR	1.2738***	0.4396
	(2.7942)	(0.8118)
Infrastructure	0.0204	0.3367***
	(0.3851)	(2.9681)
ECO	−0.1475	−0.2386
	(−1.1848)	(−1.3077)
cdev	−0.2677	0.4769
	(−1.2339)	(1.0071)
Constant	0.6064	−11.0322***
	(0.3631)	(−2.6189)
Regional FE	YES	YES
Year FE	YES	YES
N	263	264
R-squared	0.8625	0.8809

Table 5
Analysis of the impact of financial services enhancement on RUR.

VARIABLES	(1) RUR	(2) RUR	(3) RUR	(4) RUR	(5) RUR
FS	9.4715***	5.8228***	4.6582***	1.3866***	1.4233***
	(19.5969)	(11.8045)	(8.9723)	(8.5615)	(9.0206)
FIA		0.2868***	0.2742***	0.0150	0.0050
		(15.5071)	(15.1886)	(0.7260)	(0.2073)
GUP		−0.6814***	−0.5595***	−0.4877***	−1.0971***
		(−6.3592)	(−5.1677)	(−5.3754)	(−7.6622)
EduInv			0.3971	0.6512	−0.2239
			(0.7263)	(1.5111)	(−0.4803)
Upgrade			2.1536***	1.1074***	1.6067***
			(5.8472)	(3.6373)	(3.5594)
UR				−0.1003	0.6529*
				(−0.4772)	(1.9244)
Infrastructure				−0.0730***	0.1260**
				(−2.8262)	(2.4345)
ECO				−0.0988	−0.3649***
				(−1.2727)	(−4.1421)
Cdev				0.7477***	0.7721***
				(8.5093)	(7.4435)
Constant	−0.2505***	0.0372	−1.9110***	−5.2530***	−8.0463***
	(−5.5460)	(0.8387)	(−5.7824)	(−11.4912)	(−9.5220)
Regional FE	NO	NO	NO	NO	YES
Year FE	NO	NO	NO	NO	YES
N	527	527	527	527	527
R-squared	0.4614	0.6251	0.6538	0.7865	0.8018

positive effects are even more pronounced; Second, the enhancement of financial service levels demonstrates a highly significant and robust driving effect on rural revitalization; Third, the improvement of financial services plays a crucial mediating role between ECIR and RUR, revealing the intrinsic mechanism by which the two synergistically foster rural revitalization.

In light of the aforementioned research conclusions, the following policy recommendations are proposed to further advance rural revitalization and fully leverage the synergistic effects of ECIR and the enhancement of financial services: First, optimize the implementation pathway of e-commerce policies in rural areas. In regions with lower fiscal support, it is essential to increase investment in e-commerce infrastructure construction and actively attract participation from social capital; for areas with higher fiscal support, a financial fund utilization evaluation mechanism should be established to precisely support projects integrating e-commerce and industry development. Second, strengthen the construction of the rural financial service system, encourage financial institutions to

Table 6
Endogeneity test results.

VARIABLES	First stage FS	Second stage RUR
FDL	0.0172*** (19.9685)	
FS		0.1268*** (2.9260)
FIA	0.0215*** (10.0672)	0.0235 (0.6562)
GUP	0.0202*** (3.9144)	−0.2570*** (−4.0956)
EduInv	−0.0195 (−0.8129)	2.0025*** (6.1522)
Upgrade	−0.0040 (−0.3171)	−0.0273 (−0.1605)
UR	0.0050 (0.4376)	−0.6302*** (−4.1036)
Infrastructure	0.0062*** (6.1334)	−0.0918*** (−6.2759)
ECO	−0.0088** (−2.1606)	0.2250*** (4.2881)
cdev	−0.0234*** (−3.7262)	0.7270*** (7.7463)
Constant	0.1627*** (4.9392)	−6.7824*** (−13.7968)
N	527	527
LM value	226.9913	
First phase F value	379.0725	
R-squared	0.9092	0.8397

Table 7
Mediating effect analysis results of financial service enhancement.

Variables	(1) FS	(2) RUR
ECIR	0.0526*** (43.5352)	0.2857*** (3.1640)
FS		3.5141** (2.2869)
FIA	−0.0100*** (−10.7732)	0.0439 (1.2736)
GUP	−0.0083* (−1.6960)	−0.2792* (−1.6977)
EduInv	0.0186 (1.2461)	1.2665** (2.5358)
Upgrade	−0.0727*** (−5.8785)	2.3281*** (5.4507)
UR	−0.1414*** (−15.0985)	0.6851* (1.8007)
Infrastructure	0.0060*** (4.0262)	0.1026** (2.0248)
ECO	−0.0091*** (−3.5718)	−0.1381 (−1.6064)
cdev	0.0520*** (7.8115)	−0.1925 (−0.8168)
Constant	−0.4215*** (−7.9217)	−1.5726 (−0.8325)
Regional FE	YES	YES
Year FE	YES	YES
N	527	527
R-squared	0.9668	0.8504

develop customized financial products, expedite the promotion and application of financial technology in rural areas, and enhance the accessibility and suitability of financial services. Third, reinforce policy coordination and linkage by establishing a robust interdepartmental coordination mechanism that integrates resources from commerce, finance, and other sectors, promotes the sharing of e-commerce and financial service resources among regions, facilitates coordinated regional development, and comprehensively advances the level of rural revitalization.

Table 8
Analysis of bootstrap test results.

	coefficient	std. err.	z	$P>\|z\|$	[95 % conf. interval]	
_bs_1	0.0695	0.0550	5.2600	0.0060	0.0383	0.1773
_bs_2	0.0112	0.0510	3.7100	0.0120	0.0010	0.0128

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CRediT authorship contribution statement

Yifan Qi: Writing – original draft, Validation, Resources, Methodology. **Yasi Gao:** Writing – original draft, Funding acquisition, Formal analysis, Data curation. **Yanghui Wu:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Data availability

The authors do not have permission to share data.

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